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(1) Research Program/Project/Study Title: Performance of Indigenous Rice (<i>Oryza sativa</i> L. Applied with Bokashi with Different Farm Manures	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Mambusao
Province:	Capiz
Region:	VI

(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity Rice	(8) Discipline/Field of Specialization Agriculture
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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

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The study was conducted at the Organic Agriculture Area of Capiz State University – Burias Campus to evaluate the performance of indigenous rice applied with bokashi with different farm manure. The experiment was laid out using the Randomized Complete Block Design (RCBD) with two factors. Factor A considered bokashi with different farm manures such as: A0–(Control), A1– bokashi with chicken dung, A2– bokashi with sheep manure, and A3 – bokashi with hog manure and A4– commercial organic fertilizer. Factor B comprised the different varieties of indigenous rice, namely B1– red rice, B2 – black rice, B3 – Kaponpon, and B4– RC-10. Each treatment was replicated three times. Using IIRISTAR Program, with a 5% level of significance, results were analyzed. The results revealed that the growth and yield of the different indigenous rice, when applied with bokashi with different farm manures, varies significantly. The results showed that rice varieties highly significantly performed well under various treatments of bokashi with different farm manures thus resulting to an improved performance with an evident increased in plant height, biomass, length and weight of panicle, number of spike per spikelets, number of tillers per plant, number of productive tillers, unfilled grains and 1000 seeds weight. It further proves that bokashi with different farm manures highly significantly affected the length and weight of the panicle, number of spikes per spikelet, 1000 seeds weight, and an improved grain yield. Among the parameters, only the 1000 seeds’ weight has an interaction effect. Results of this study corroborate that bokashi with different farm manures, when applied in different rice varieties, is an effective organic fertilizer.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Keywords: bokashi, manure, indigenous rice, organic fertilizer

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Although rice is one of the staple foods in the Philippines, it has its own staple problems too. As production costs remain high due to costly fertilizer and other farm inputs and as local harvests continue to drop, the government's ambition to lower rice prices has proven difficult. Other hurdles may include natural calamities, pests, farm-to-market roads, technical know-how and farm management practices.

Today, organic farming has now received global prominence. In fact, the Department of Agriculture (DA), encourages farmers to shift from chemical-based farming to organic, saying, “organic agriculture is a key element in moving forward” (Da-Afid, 2022). The adoption rate is still low despite the Philippine government's efforts to foster the cultivation of organic rice for a multitude of reasons.

For instance, a study by Pantoja et. al (2016) determined that most of the farmer-respondents did not switch to organic farming because organic farming decreases yield, poorly taught information about organic farming that led to misconceptions and the perception that organic farming required more time, especially for farm workers and owners. Other factors included a decline in profit, difficult cultivation, and adherence to traditional rice-growing methods.

Nonetheless, a study by Pornpratansombat P., et al. (2011) supported that farmers concur on chemical-free agriculture methods, and the continued use of conventional farming techniques results in problems on farms, whereas organic farming is seen to provide a viable alternative to traditional

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farming challenges. Organic fertilizer, through its impact on the soil's physical, chemical, and biological qualities, can nourish the soil with its nutrients, including bokashi.

Bokashi is a Japanese word meaning "fermented organic matter" (Vanderlinden, 2022 Bokashi may physically loosen the soil and elevate the pH of the soil, both of which will expand the root space. For plant roots, access to nutrients becomes greater. Biologically, bokashi can boost the quantity of synthetic and fermented microorganisms, preventing the spread of illness and insect harm (Making of Bokashi Fertilizer From Rice Straw (*Oryza Sativa* L.) by Using the Activator Effective Microorganisms (EM4), 2020).

The study of Abd Hamid, and Muhammad Nabil (2019) to see the extent to which using bokashi affected the growth and development of rice farming revealed that plant height, number of tillers, number of leaves, straw biomass, root biomass, and root length showed significantly when applied with bokashi, and thus, enhanced rice cultivation's performance during the vegetative phase.

Consequently, the height of rice plant, number of leaves, tillers, and shoot/root ratio of the plant were best affected by the treatment of bokashi bottom ash. In contrast, the other application enhanced the number of productive tillers, the quantity of filled grains, and the weight of the grains (Sondari & Nurkhalidah, 2012).

The study intends to establish if combining bokashi infused with different animal manures (chicken dung, hog and sheep) increases the performance or growth of indigenous rice relative to nutriboost fertilizer.

Objective of the Study

The study was conducted to determine the performance of indigenous rice varieties applied with bokashi with different farm manures. Specifically, the study aims to;

1. determine the growth and yield performance of indigenous rice varieties of indigenous rice applied with bokashi with different farm manures;
2. best substrates in growing different indigenous rice varieties;
3. interaction effect between the different rice varieties and bokashi with different farm manures.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

This study was conducted to evaluate the performance of indigenous rice applied with bokashi with different farm manure. It was laid out using the Randomized Complete Block Design (RCBD) with two factors. Factor A consisted bokashi with different farm manures such as: A0 - Control, A1 - Bokashi with chicken dung, A2 – bokashi with sheep manure, A3 – hog manure, and A4 – Commercial Fertilizer (COF). On the other hand, Factor B comprised the different varieties of indigenous rice, namely: B1 – Red rice, B2 – Black rice, B3 – Kaponpon rice, and B4 – RC - 10 rice. Each treatment was replicated three times. All data gathered were analyzed through the IRRISTAR Program and results were interpreted at 5%

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level of significance. Differences among treatments was determined using the Duncan Multiple Range Test (DMRT) and were interpreted at 5% level of significance.

(12) **Results and Discussions** (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Plant Height (cm)

The ANOVA of the plant height revealed a highly significant results for rice varieties. However, there is no significant effect of bokashi with different farm manures.

On different indigenous rice varieties, black rice obtained the tallest plant height with a mean of 119.19 cm, which was found comparable with RC -10 (115.84 cm), and red rice (110.61 cm) while the Kaponpon variety produced the shortest plant height (66.93 cm). Regarding the effect of bokashi on different farm manures, the means ranged from 101.53 cm to 107.13 cm.

There was no interaction effect between the varieties of the different indigenous rice and bokashi with different farm manures.

Biomass (grams)

The ANOVA for the biomass showed a highly significant effect on the different varieties of indigenous rice but not a significant effect on bokashi with different farm manures. In terms of bokashi with different farm manures, the means ranged from 79.36 g. to 92.97 g. For the different indigenous rice varieties, RC-10 attained the heaviest biomass with a mean of 115.68 g., which was followed by black rice (89.71 g.), which was found comparable with red rice (86.80 g.). The results further revealed that the Kaponpon variety obtained the lightest biomass with a mean of 59.22 grams.

There was no interaction effect between the varieties of the different indigenous rice and bokashi with different animal manures. Thus, this did not affect the biomass of rice.

Number of Leaves

The analysis for the number of leaves revealed that there was a significant effect on the different indigenous rice varieties but not on bokashi with different farm manures.

In terms of the effect of different varieties of indigenous rice, black rice has the greatest number of leaves (34.65) and was found to be comparable with that of the Kaponpon variety (34.15) and Red rice (31.64). RC – 10 variety produce the least number of leaves which was found comparable with that of Red rice (31.64). On the other hand, bokashi with different farm

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manures the means ranged from 31.70 to 31.57 number of leaves.

Moreover, there was no interaction effect between the variety of indigenous rice leaves and the bokashi with different farm manures.

Stem girth (mm)

Analysis of variance revealed that there was a significant effect on the different indigenous rice varieties but not in the bokashi with different farm manures. In terms of different indigenous rice varieties, black rice variety was observed to have the thickest stem girth (20.91mm) which was found to be comparable with the RC10 variety (19.36 mm) and Red rice variety (18.63 mm). While the Kaponpon variety has the thinnest width (17.06 mm) among all the rice varieties which also found to be comparable from that of Red rice (18.63 mm), and RC – 10 (19.36 mm). In terms of bokashi with different farm manures, the means ranged from 18.12 mm to 20.87 mm.

Nonetheless, there was no significant difference between the stem girth of indigenous rice varieties and bokashi with different farm manures.

Number of Tillers per Plant

The analysis of variance for the number of tillers revealed that there was a significant effect on the different indigenous rice varieties but not on bokashi with different farm manures. In terms of bokashi with different farm manures, the means ranged from 7.99 to 8.62 tillers. In terms on the different indigenous rice varieties, black rice has the greatest number of tillers with a mean of 9.18, and was found to be comparable with that of the Kaponpon variety (8.99) and red rice (8.60). In contrast, RC -10 obtained the least number of tillers per plant (6.66).

However, there was no significant difference between the number of tillers per plant on the different indigenous rice varieties and bokashi with different farm manures.

Number of Productive Tillers

The analysis of variance for the number of productive tillers revealed a highly significant effect on the different indigenous rice varieties and no significant effect on the application of bokashi with different farm manures. For the effect of bokashi with different farm manures, the means ranged from 6.02 to 6.49 tillers.

In terms of indigenous rice variety, black rice had the greatest number of productive tillers with a mean of 7.83, which was found comparable with that of the red rice(6.65); this was being followed by the Kaponpon variety (6.45) which was also found to be comparable from that of red rice (6.65), while the RC-10 has the least number of tillers (4.44).

ANOVA further reflected no significant interaction effect between the two factors. This signifies that the indigenous rice varieties when applied with bokashi with different farm manures did not interact in influencing the number of productive tillers of the different indigenous rice varieties.

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Number of Non-Productive Tillers

Analysis revealed that the number of non-productive tillers was not significantly affected by the different indigenous rice varieties and bokashi with different farm manures. In terms of bokashi with different farm manures, the means ranged from 2.71 to 2.79 in numbers. On the other hand, for the different indigenous rice varieties, the means ranged from 2.43 to 2.59.

However, there was no significant interaction between the number of non-productive tillers of the different indigenous rice varieties and bokashi with different farm manures.

Length of Panicle (cm)

The panicle length of the different varieties of indigenous rice was highly significantly affected both by the application of bokashi with different farm manures and the rice varieties. However, it was noted that there was no interaction effect between the two factors.

Among the different indigenous rice varieties, RC10 variety displayed the longest panicle length (27.14 cm.), which was statistically followed by red rice (21.56 cm.), black rice (20.64 cm.), while Kaponpon variety displayed the shortest length of panicle(19.39).

On the one hand, among bokashi with different farm manures, bokashi with sheep manure revealed the highest mean (22.78), which was found comparable with that of bokashi with chicken dung (22.53) in which likewise statistically comparable with bokashi with hog manure (22.13). This was followed by bokashi with chicken dung (22.53), which was also found to be comparable with that of bokashi with hog manure (22.13) and commercial organic fertilizer (21.87). The shortest panicle was obtained by rice varieties with no application of bokashi (21.61), which was also found to be comparable with the length of panicle from rice varieties applied with commercial organic fertilizer (21.87) and bokashi with hog manure (22.13).

Weight of Panicle (g)

The ANOVA for the weight of the panicle revealed a highly significant effect both for the different indigenous rice varieties and bokashi with different farm manures.

For the different indigenous rice varieties, RC–10 produced the heaviest panicle with a mean of 4.01 grams, followed by Red rice (3.13 g), Black rice (2.83 g), while the Kaponpon had the lightest weight of panicle (2.54 g).

In terms of bokashi with different farm manures, bokashi with hog manure produced the heaviest weight of panicle with a mean of 3.36 g., which was also found to be comparable with that of bokashi with sheep manure (3.33 g) and bokashi with chicken dung (3.23 g).

Similar ANOVA results revealed that the bokashi with hog manure (3.23 g) was also found comparable with the application of commercial organic fertilizer. Meanwhile, the lightest weight was recorded from plants with no application (2.80 g.), which was also found to be comparable with that of commercial organic fertilizer (2.92 g.).

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No interaction effect between the different indigenous rice varieties and bokashi with different farm manures was reflected.

Number of Filled Grains

The analysis of variance for the number of filled grains revealed a highly significant effect on the different indigenous rice. However, no significant effect was found in the application of bokashi with different farm manures.

In terms of bokashi with different farm manures, the means ranged from 105 to 116. In terms of different rice varieties, RC-10 obtained the most number of filled grains (151.95), followed by red rice (108.10), and the Kaponpon variety (93.67) filled grains. The least number of filled grains was obtained by the black rice (85.5).

ANOVA further indicated that there was no significant interaction effect between the bokashi with different farm manures and different indigenous rice. This implies that the two factors did not interact in influencing the number filled grains of the different indigenous rice varieties.

Number of Unfilled Grains

The analysis of variance revealed that the number of unfilled grains was highly significantly affected by different rice varieties but not with the effect of bokashi with different farm manures. In terms of the effect of bokashi with different farm manures, the mean ranged from 33.51 to 38.83 unfilled grains.

RC-10 produced the greatest number of unfilled grains with a mean of 60.73, followed by black rice (29.65), which was found statistically comparable with red rice (29.91).

No interaction effect was found.

Number of Spikelets per Panicle

Analysis of variance revealed that the number of spikelets per panicle was highly significantly affected by the indigenous rice varieties but not in the application of bokashi with different farm manure. The means ranged from 26.09 to 31.80 spikelets per panicle.

The highest number of spikelets per panicle, in terms of different rice varieties, was obtained by the RC10 variety with a mean of 43.32, followed by the red rice (26.14), which was also found to be comparable with that of black rice (23.62), while Kaponpon variety had the least number of spikelets per panicle (18.37).

The analysis further revealed that the two factors did not influence the number of spikelets per panicle of the different indigenous rice varieties applied with bokashi with different farm manures.

Number of Spike per Spikelets

The ANOVA for the number of spikes per spikelets revealed a highly significant effect on the different indigenous rice and a significant effect in the application of bokashi with different farm manures.

Rice varieties applied with bokashi with hog manure produces the most number of

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spike per spikelets with a mean of 10.13 spikes and it was found to be comparable to rice varieties applied with commercial organic fertilizer (10.04), bokashi with chicken dung (9.98), and bokashi with sheep manure (9.92), while the least number of spikes was found in rice varieties with non- application of bokashi with different farm manures.

For the effect of different rice varieties, RC-10 (151.95) had the most number of spike per spikelets (13.71), followed by red rice (9.14), and Kaponpon variety (8.69). The least number of spike per spikelets was obtained by the black rice with a mean of 7.99.

ANOVA further indicated that there was no significant interaction effect between the bokashi with different farm manures and different indigenous rice. Hence, the two factor did not interact in influencing the number of spike per spikelets of the different indigenous rice varieties.

1000 Seeds weight (g)

The analysis for 1000 seeds weight showed that there was a highly significant effect on the different indigenous rice varieties and with the application of bokashi with different farm manures. An interaction effect between the two factors was also reflected in the results of ANOVA.

On the bokashi with farm manures, the mean ranged from 23.86g to 27.27g. On the other hand, the heaviest 1000 seeds weight on different indigenous rice varieties was obtained by the Kaponpon variety with a mean of 28.70 grams, which was found comparable with red rice (27.86 g), followed by black rice (25.55), while RC 10 obtained the lightest 1000 seeds weight (21.24).

Application of bokashi with hog manure, in Kaponpon variety, yielded the heaviest 1000 seeds weight with a mean of 29.50 grams and was statistically found comparable with red rice, applied with bokashi with hog manure, Kaponpon variety (29.12 g.) with no application, Kaponpon variety (28.72 g.) applied with bokashi with sheep manure, red rice, and Kaponpon variety applied with bokashi with chicken dung, Kaponpon variety (28.07 g.) and red rice (27.96 g.) applied with commercial organic fertilizer, red rice (27.73 g) applied with bokashi with sheep manure, RC-10 (27.63 g) with no application, black rice applied with bokashi with sheep and hog manure and with no application.

It was then followed by red rice where there was no treatment applied. It was found to be comparable to red rice applied with bokashi with sheep manure, RC – 10 (20.86 g), applied with bokashi with hog manure, black rice (19.62 g), and Kaponpon variety (19.62 g) applied with bokashi with chicken dung.

The lightest 1000 seeds weight was obtained by RC-10 variety (18.39 g.) applied with bokashi with commercial organic, which was found to be comparable with that of RC-10 varieties applied bokashi with hog manure (19.68 g) and sheep manure (19.62 g).

Grain Yield

ANOVA statistical results for grain yield in tons per hectare revealed a highly significant effect in the application of bokashi with different farm manures and no significant effect on different rice varieties of which means ranged from 0.07 to 0.08 tons/ha.

For the application of the bokashi with different farm manures, bokashi with sheep

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manure has the highest mean (0.09 tons/ha), which was found to be comparable with that of the hog manure (0.08 tons/ha), in which hog manure was also found to be comparable with that of chicken dung (0.08 tons/ha). Furthermore, commercial organic fertilizer and non-application of bokashi with different farm manures recorded the lowest grain yield in ton per hectare (0.07), which was found to be comparable with that of bokashi with chicken dung (0.08 tons/ha). Furthermore, no interaction effect was noted in the result of ANOVA.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

CONCLUSIONS

Based on the results of the study, the following conclusions were drawn:

1. Growth and yield of the different indigenous rice, when applied with bokashi with different farm manures, varies significantly.
2. Indigenous rice varieties had a highly significant effect on the growth and yield of all the parameters gathered except for the number of non-productive tillers, and grain yield. It also showed a significant effect on the number of leaves and spikelets .
3. Bokashi with different farm manures is an effective organic fertilizer as shown to be highly significantly effective on the length and weight of the panicle, number of spike per spikelets, and 1000 seeds weight but not in other parameters.
4. All the different substrates utilized in making bokashi demonstrated a comparable result.
5. Results of this study revealed that among all the parameters, the only interaction effect was found in the 1000 seeds weight.

RECOMMENDATIONS

Based on the results of this study, the following recommendations are hereby forwarded:

1. Grow either red rice, black rice or RC-10 rice for higher grain yield when using Bokashi with different farm manures as fertilizers.
2. Use any animal manure, such as sheep, hog, and chicken manure in making bokashi.
3. Analyze the physical, chemical, and microbial properties of the soil before and after cropping.
4. Validate and conduct a similar study for further evaluation.

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(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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Table 1. Growth and Yield Performance of Rice (*Oryza sativa* L.) as Affected by Bokashi with Different Farm Manures

	Plant Height (cm)						Biomass					
Treatment	Varieties**				Total	Mean	Varieties**				Total	Mean
Bokashi	B1- Red rice	B2- Black rice	B3- Kaponpon	B4- RC-10			B1- Red rice	B2- Black rice	B3- Kaponpon	B4- RC-10		
A- Control	109.88	113.06	66.68	116.5	406.12	101.53	78.2	81.8	52	105.43	317.43	79.36
B- Bokashi with Chicken Dung	112.88	115.35	69.82	120.17	418.22	104.55	97.3	87.1	72.5	105.33	362.23	90.56
C- Bokashi with Sheep manure	107.62	116.92	72.51	116	413.05	103.26	82.6	97.93	62.47	107.5	350.5	87.63
D- Bokashi with Hog manure	109.15	103.52	70.02	114.22	396.91	99.23	79.8	91.63	59.03	124.57	355.03	88.76
E- Commercial Organic	113.51	147.11	55.60	112.31	428.53	107.13	96.1	90.1	50.10	135.57	371.87	92.97
Total	553.03	595.96	334.63	579.20			434	448.56	296.1	578.4		
Mean	110.61a	119.19a	66.93b	115.84a			86.80b	89.71b	59.22c	115.68a		

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	Number of Leaves						Stem Girth					
Treatment	Varieties*				Total	Mean	Varieties*				Total	Mean
Bokashi	B1- Red rice	B2- Black rice	B3- Kaponpon	B4- RC-10			B1- Red rice	B2- Black rice	B3- Kaponpon	B4- RC-10		
A- Control	33.83	35	29.87	28.1	126.8	31.70	18.87	28.5	17.27	18.83	83.47	20.87
B- Bokashi with Chicken Dung	32.33	30.2	42.4	23.73	128.66	32.17	18.23	22.73	16.9	19.43	77.29	19.32
C- Bokashi with Sheep manure	28.87	40.23	34.1	24.1	127.3	31.83	19	17.87	17.57	19.33	73.77	18.44
D- Bokashi with Hog manure	29.83	32.43	30.5	38.3	131.06	32.77	18.37	17.33	17.23	19.9	72.83	18.21
E- Commercial Organic	33.33	35.4	33.87	23.67	126.27	31.57	18.7	18.13	16.33	19.3	72.46	18.12
Total	158.19	173.26	170.74	137.9			93.17	104.56	85.3	96.79		
Mean	31.64ab	34.65a	34.15a	27.58b			18.63ab	20.91a	17.06b	19.36ab		

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	Numbers of tiller per Plant						Number of Productive Tillers					
Treatment	Varieties*				Total	Mean	Varieties**				Total	Mean
Bokashi	B1- Red rice	B2- Black rice	B3- Kaponpon	B4- RC-10			B1- Red rice	B2- Black rice	B3- Kaponpon	B4- RC-10		
A- Control	9.5	9.17	8.57	6.93	34.17	8.54	6.93	8.83	5.53	4.67	25.96	6.49
B- Bokashi with Chicken Dung	8.73	8.23	10.93	6.6	34.49	8.62	7	6.87	7.53	4.4	25.8	6.45
C- Bokashi with Sheep manure	8.1	10.13	8.7	6.6	33.53	8.38	5.97	8.83	6.67	4.47	25.94	6.49
D- Bokashi with Hog manure	7.63	8.77	8.3	7.27	31.97	7.99	6.07	6.57	6.43	5	24.07	6.02
E- Commercial Organic	9.03	9.60	8.43	5.9	32.96	8.24	7.3	8.07	6.10	3.67	25.14	6.29
Total	42.99	45.9	44.93	33.3			33.27	39.17	32.26	22.21		
Mean	8.60a	9.18a	8.99a	6.66b			6.65ab	7.83a	6.45b	4.44c		

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	Number of non-Productive Tillers ^{ns}						Panicle Length					
Treatment	Varieties				Total	Mean	Varieties**				Total	Mean
Bokashi	B1- Red rice	B2- Black rice	B3- Kaponpon	B4- RC-10			B1- Red rice	B2- Black rice	B3- Kaponpon	B4- RC-10		
A- Control	2.66	2.77	3.14	2.57	11.14	2.79	20.52	20.64	19.49	25.78	86.43	21.61c
B- Bokashi with Chicken Dung	2.63	1.99	3.54	2.63	10.79	2.70	22.24	20.44	19.4	28.04	90.12	22.53ab
C- Bokashi with Sheep manure	2.38	2.66	2.45	2.32	9.81	2.45	22.48	21.36	19.54	27.75	91.13	22.78a
D- Bokashi with Hog manure	2	2.47	2.29	2.94	9.7	2.43	21.6	19.75	19.31	27.84	88.5	22.13c
E- Commercial Organic	2.5	2.72	3.11	2.51	10.84	2.71	20.95	21.03	19.19	26.29	87.46	21.87bc
Total	12.17	12.61	14.53	12.97			107.79	103.22	96.93	135.7		
Mean	2.43	2.52	2.91	2.59			21.56b	20.64c	19.39d	27.14a		

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	Weight of Panicle						Number of Filled grains					
Treatment	Varieties**				Total	Mean	Varieties**				Total	Mean
Bokashi**	B1- Red rice	B2- Black rice	B3- Kaponpon	B4- RC-10			B1- Red rice	B2- Black rice	B3- Kaponpon	B4- RC-10		
A- Control	2.77	2.73	2.25	3.44	11.19	2.80	92.3	88.77	91.87	147.5	420.44	105.11
B- Bokashi with Chicken Dung	3.38	2.74	2.58	4.23	12.93	3.23	120.5	91.87	91.87	160.3	464.54	116.14
C- Bokashi with Sheep manure	3.14	3.13	2.67	4.36	13.3	3.33	110.63	87.7	91.8	151.13	441.26	110.32
D- Bokashi with Hog manure	3.57	2.63	2.79	4.44	13.43	3.36	107.2	75.53	100.47	154.47	437.67	109.42
E- Commercial Organic	2.78	2.93	2.39	3.57	11.67	2.92	109.87	83.67	92.33	146.37	432.24	108.06
Total	15.64	14.16	12.68	20.04			540.5	427.54	468.34	759.77		
Mean	3.13b	2.83c	2.54d	4.01a			108.10b	85.51d	93.67c	151.95a		

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	Number of UnFilled grains						Number of spikelets per panicle					
Treatment	Varieties**				Total	Mean	Varieties**				Total	Mean
Bokashi	B1- Red rice	B2- Black rice	B3- Kaponpon	B4- RC-10			B1- Red rice	B2- Black rice	B3- Kaponpon	B4- RC-10		
A- Control	33.13	29.43	25.73	67.03	155.32	38.83	23.03	24.43	18.07	38.83	104.36	26.09
B- Bokashi with Chicken Dung	35.07	20.13	19.5	59.33	134.03	33.51	29.8	19.27	15.03	42.7	106.8	26.70
C- Bokashi with Sheep manure	26.03	39.47	27.07	57.6	150.17	37.54	26.7	25.67	19.43	55.4	127.2	31.80
D- Bokashi with Hog manure	27.83	25.1	24	63.97	140.9	35.23	24.17	20.4	20.3	42.87	107.74	26.94
E- Commercial Organic	27.47	34.10	30.60	55.7	147.87	36.97	27	28.33	19.03	36.4	110.76	27.69
Total	149.53	148.23	126.9	303.63			130.7	118.1	91.86	216.2		
Mean	29.91b	29.65b	25.38b	60.73a			26.14b	23.62b	18.37c	43.24a		

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	Number of Spike per spikelets						1000 seeds weight					
Treatment	Varieties**				Total	Mean	Varieties**				Total	Mean
Bokashi	B1- Red rice	B2- Black rice	B3- Kaponpon	B4- RC-10			B1- Red rice	B2- Black rice	B3- Kaponpon	B4- RC-10		
A- Control	8.4	8.1	8.5	12.47	37.47	9.37b	26.3	26.43	29.12	27.63	109.48	27.37
B- Bokashi with Chicken Dung	9.6	7.73	8.47	14.1	39.9	9.98a	28.1	19.62	28.1	19.62	95.44	23.86
C- Bokashi with Sheep manure	9.17	8.07	8.73	13.7	39.67	9.92a	27.73	27.48	28.72	20.86	104.79	26.20
D- Bokashi with Hog manure	9.1	7.8	9.03	14.57	40.5	10.13a	29.19	27.34	29.5	19.68	105.71	26.43
E- Commercial Organic	9.43	8.27	8.73	13.73	40.16	10.04a	27.96	26.89	28.07	18.39	101.31	25.33
Total	45.7	39.97	43.46	68.57			139.28	127.76	143.51	106.18		
Mean	9.14 b	7.99d	8.69c	13.71a			27.86a	25.55b	28.70a	21.24c		

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	Grain Yield in tons/ha					
Treatment	Varieties				Total	Mean
Bokashi**	B1- Red rice	B2- Black rice	B3- Kaponpon	B4- RC-10		
A- Control	0.07	0.07	0.08	0.07	0.29	0.07c
B- Bokashi with Chicken Dung	0.08	0.08	0.07	0.07	0.3	0.08c
C- Bokashi with Sheep manure	0.08	0.11	0.08	0.07	0.34	0.09a
D- Bokashi with Hog manure	0.08	0.09	0.07	0.08	0.32	0.08ab
E- Commercial Organic	0.07	0.07	0.06	0.06	0.26	0.07c
Total	0.38	0.42	0.36	0.35		
Mean	0.08	0.08	0.07	0.07		